

ARITRI HALDER

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Mechanical Engineer | Thermal Systems | GD&T | Prototype Builds | Test & Validation | NPD/NPI

Summary

Ph.D. candidate in Mechanical Engineering with hands-on R&D experience in DFMA, V&V, and NPD. Designed and tested a novel 20 kW high-temperature industrial burner, integrated sensors and controls, validated system performance, and used experimental data to guide engineering decisions, achieving a 60% reduction in NOx emissions compared with conventional burners.

Education

University of Minnesota - Twin Cities

Ph.D., Mechanical Engineering

M.S., Mechanical Engineering

Minneapolis, MN

Expected June 2027 | GPA: 3.66/4.00

June 2025 | GPA: 3.61/4.00

Jadavpur University

B.E., Mechanical Engineering

Kolkata, India

July 2023 | GPA: 9.12/10.00

Work Experience

University of Minnesota Twin Cities - Graduate Research Assistant

Minneapolis, MN | Oct 2023 - Present

Process Equipment Development & Test Infrastructure

- Designed and validated a novel 20 kW high-temperature burner to enable stable 100% ammonia operation, achieving up to 60% lower NOx emissions than conventional burners.
- Developed SolidWorks CAD models, GD&T drawings, and BOMs of swirl vanes and nozzle geometry to maintain symmetric flow and improve combustion performance.
- Selected stainless-steel wetted components and PTFE seals for ammonia compatibility, enabling safe fuel delivery while supporting high temperature operation near 1000 deg C.
- Designed modular, interchangeable burner components so swirl and jet geometries could be changed independently, accelerating prototype iteration without rebuilding the test platform.
- Planned and executed a four-variable DOE to map stable operating windows and relate operating changes to NOx, ammonia slip, and flame behavior, guiding design decisions.
- Engineered and commissioned a three-stream ammonia/methane/air system with mass flow controls, sensors, NI cDAQ, and LabVIEW, enabling automated flow control and synchronized data acquisition.
- Diagnosed failures across gas delivery, controls, instrumentation, and diagnostics using RCA/FMEA; implemented corrective actions that restored repeatable testing and improved operational safety.
- Collaborated with CenterPoint Energy and University of Minnesota Duluth to convert validation data into design recommendations supporting scale-up for industrial thermal applications.

Instrumentation, Optical Diagnostics & Characterization

- Built and aligned a 2D-2C PIV optical system using convex, concave, and cylindrical optics; optimized the laser sheet to quantify velocity and recirculation structures.
- Designed and validated a rotational camera mount for repeatable 20-angle flame imaging, enabling 3D chemiluminescence reconstruction with a single camera.
- Integrated advanced laser and optic diagnostics such as FTIR, PIV, chemiluminescence, and tomography measurements to connect emissions with flow and flame structure, identifying operating conditions that reduced NOx.
- Developed custom image processing algorithms in Python/MATLAB workflows to process 200-300 images per condition, and quantify burner process behavior and iterate design improvements.

Jadavpur University - Research Assistant

Kolkata, India | Aug 2022 - Jun 2023

- Developed parallel C/C++ immersed-boundary simulations on an HPC platform to resolve fluid-structure interaction at $Re = 250$, enabling systematic evaluation of unsteady wake response.
- Varied Strouhal number and pitching phase to determine how structural kinematics altered vortex shedding and flow separation, mapping wake transitions using Tecplot.

IIT Kanpur - Research Intern

Kanpur, India | May 2022 - Jul 2022

- Established mesh-independent ANSYS Fluent models before comparing insert designs, ensuring predicted pressure-drop and heat-transfer trends were independent of grid resolution.
- Used the validated CFD model to optimize 5 mm and 10 mm insert geometries across $Re = 200-900$, improving thermal-fluid performance by up to 28% over the baseline design.

IIT Madras - Research Assistant

Chennai, India | May 2022 - Sep 2022

- Modeled bio-inspired sharkskin surfaces with an immersed-boundary finite-difference solver to handle complex geometry, quantifying effects on skin friction and boundary-layer transition.
- Applied PSD, scalograms, and Strouhal-number scaling to velocity fluctuations to identify dominant frequencies and relate them to surface-induced unsteady flow behavior.

Technical Skills

Mechanical Design & Manufacturing: SolidWorks 3D CAD, mechanical parts and subsystems, engineering drawings, GD&T, DFM/DFA, tolerance stack-up, fit/clearance analysis, BOMs, machining, assembly, materials selection, prototyping, NPD/NPI.

Thermal-Fluid & Engineering Analysis: ANSYS Fluent, ANSYS DesignModeler, CFD, thermal analysis, heat transfer, thermodynamics, fluid mechanics, FSI, mesh convergence, numerical methods, HPC, Cantera.

Experimental Systems & Diagnostics: PIV, chemiluminescence, computed tomography of chemiluminescence, FTIR spectroscopy, optical diagnostics, camera calibration, laser diagnostics, emissions characterization.

Instrumentation, Controls & Validation: NI cDAQ/DAQ, LabVIEW, MFCs, pressure/temperature sensors, system integration, equipment commissioning, V&V, DOE, RCA, FMEA, calibration, test automation.

Programming & Data Analysis: Python, MATLAB, C/C++, Excel, Tecplot, image processing, statistical analysis, automated data processing, data visualization, technical reporting, GitHub.

Awards & Professional Activities

- Chester E. Dekko Foundation Endowed Graduate Fellowship, University of Minnesota, 2024-2025; Graduate Student Fellowship, Department of Mechanical Engineering.
- Best Abstract, Thermal Applications, 4th Symposium on Ammonia Energy, 2025.
- Served as a peer reviewer in Combustion Science and Technology.
- Social Chair, Mechanical Engineering Graduate Council, University of Minnesota, 2024-2026; organized departmental events supporting graduate-student engagement.
- Graduated from Jadavpur University with First Class Distinction.

Teaching & Mentoring Experience

Teaching Assistant, Department of Mechanical Engineering, University of Minnesota 2024, 2025, 2026

- ME 4431 Energy Conversion Laboratory:** Mentored and supervised 15+ undergraduate students in experimental procedures, instrumentation, data acquisition, and engineering analysis.
- ME 3333 Heat Transfer:** Taught undergraduate students conduction, convection, radiation, thermal-resistance networks, and engineering problem solving.

Publications and Presentations

Publications

- A. Halder**, C. Goertemiller, A. Hoxie, and W. F. Northrop, "Experimental investigation of a 20 kW atmospheric pressure pure ammonia jet-swirl burner," *Proceedings of the Combustion Institute*, vol. 42, Art. no. 106019, 2026. [\[Link\]](#).
- A. Halder**, R. Pandey, and U. Madanan, "Computer simulations on heat transfer enhancement in a minichannel heat sink using modified butterfly inserts," in *Proceedings of the 27th National and 5th International ISHMT-ASTFE Heat and Mass Transfer Conference*, pp. 149–154, 2024. [\[Link\]](#).

Selected Presentations

Five oral/poster presentations at international and U.S. technical meetings, including the 41st International Symposium on Combustion (Kyoto, Japan, 2026), 14th U.S. National Combustion Meeting (Boston, MA, 2025), and 4th Symposium on Ammonia Energy (2025).